

Tutorial 6: Multiple Regression + Inference (Chapter 3 C2 & Chapter 4 C1)

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Contents

Overview	2
1. Supporting tools (quick)	2
1.1 VIF (Variance Inflation Factor)	2
1.2 Linear hypothesis tests (F-tests)	2
2. Chapter 3 — Computer Exercise C2 (HPRICE1)	3
2.1 What the problem is asking	3
2.2 Step-by-step plan (how to solve it)	3
2.3 Common pitfalls	3
2.4 Solution in R	4
(ii) One more bedroom (holding sqrft constant)	5
(iii) One more bedroom + 140 square feet	5
(iv) Percent variation explained (R-squared)	5
(v) Predicted selling price for the first house	5
(vi) Residual for the first house; underpaid or overpaid?	6
3. Chapter 4 — Computer Exercise C1 (VOTE1)	6
3.1 What the problem is asking (plain English)	6
3.2 Step-by-step plan (how to solve it)	6
3.3 Common pitfalls	6
3.4 Solution in R	6
(ii) State and test the offset null: $\beta_1 + \beta_2 = 0$	7
(iv) Reparameterize to get the t statistic directly	8
Wrap-up	8

Overview

This tutorial covers:

- **Chapter 3, Computer Exercise C2 (HPRICE1):** multiple regression, prediction, and residuals.
- **Chapter 4, Computer Exercise C1 (VOTE1):** log-level model and testing a linear restriction.
- Two supporting tools (used in the original script):
 - VIF (variance inflation factor),
 - Linear hypothesis testing (F tests).

We will use the pipe operator `%>%` (from `dplyr`) for readability.

```
if (!requireNamespace("wooldridge", quietly = TRUE)) install.packages("wooldridge")
if (!requireNamespace("dplyr", quietly = TRUE)) install.packages("dplyr")
if (!requireNamespace("car", quietly = TRUE)) install.packages("car")

library(wooldridge)
library(dplyr)
library(car)
```

1. Supporting tools (quick)

1.1 VIF (Variance Inflation Factor)

VIF for regressor j is:

$$VIF_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is from regressing x_j on the other regressors.

```
data("wage1")

lm_wage1 <- lm(log(wage) ~ educ + exper + tenure, data = wage1)

# VIFs
vif(lm_wage1)

##      educ      exper      tenure
## 1.112771 1.477618 1.349296

# Example: compute R^2_j for educ and confirm VIF formula
r2_educ <- summary(lm(educ ~ exper + tenure, data = wage1))$r.squared
c(R2_educ = r2_educ, VIF_educ = 1/(1 - r2_educ))

##      R2_educ      VIF_educ
## 0.1013423 1.1127708
```

1.2 Linear hypothesis tests (F-tests)

A joint hypothesis like

$$H_0 : \beta_1 = 0, \beta_2 = 0, \beta_3 = 0$$

is tested using an F-test.

```
linearHypothesis(lm_wage1, c("educ = 0", "exper = 0", "tenure = 0"))
```

```
## Linear hypothesis test
##
## Hypothesis:
## educ = 0
## exper = 0
## tenure = 0
##
## Model 1: restricted model
## Model 2: log(wage) ~ educ + exper + tenure
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      525 148.33
## 2      522 101.46   3    46.874 80.391 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2. Chapter 3 — Computer Exercise C2 (HPRICE1)

2.1 What the problem is asking

We estimate:

$$price = \beta_0 + \beta_1 \text{sqft} + \beta_2 \text{bdrms} + u,$$

where **price is measured in thousands of dollars**.

Tasks: 1. Estimate the regression and write it in equation form. 2. Interpret $\hat{\beta}_2$: one more bedroom holding sqft constant. 3. Compute the predicted price change for adding a bedroom **and** 140 square feet. 4. Report the percent variation in price explained by sqft and bdrms (use R^2). 5. Predict the selling price for the first house in the sample. 6. Compute the residual for the first house and interpret underpaid vs overpaid.

2.2 Step-by-step plan (how to solve it)

- 1) Load `hprice1` and run `lm(price ~ sqft + bdrms)`.
- 2) Extract coefficients, standard errors, n, and R^2 .
- 3) Keep track of units: changes in **price** are in **thousands of dollars** (multiply by 1000 for dollars).
- 4) Use linear prediction:
$$\Delta \hat{price} = 140 \hat{\beta}_1 + 1 \cdot \hat{\beta}_2.$$
- 5) Use `predict()` for the first house. Residual = actual – predicted.

2.3 Common pitfalls

- **Units mistake:** price is in thousands of dollars.
- **R-squared choice:** the question asks “percent explained,” which is usually R^2 , not adjusted R^2 .
- **Interpretation slip:** $\hat{\beta}_2$ is “one more bedroom holding sqft fixed.”

2.4 Solution in R

```
data("hprice1")
hprice1 <- as_tibble(hprice1)

hprice1 %>% select(price, sqrft, bdrms) %>% summary()
```

```
##      price      sqrft      bdrms
## Min.   :111.0   Min.   :1171   Min.    :2.000
## 1st Qu.:230.0   1st Qu.:1660   1st Qu.:3.000
## Median :265.5   Median :1845   Median :3.000
## Mean   :293.5   Mean    :2014   Mean    :3.568
## 3rd Qu.:326.2   3rd Qu.:2227   3rd Qu.:4.000
## Max.    :725.0   Max.    :3880   Max.    :7.000
```

```
fit_c2 <- lm(price ~ sqrft + bdrms, data = hprice1)
summary(fit_c2)
```

```
##
## Call:
## lm(formula = price ~ sqrft + bdrms, data = hprice1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -127.627  -42.876   -7.051   32.589  229.003
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -19.31500    31.04662  -0.622   0.536
## sqrft        0.12844     0.01382   9.291 1.39e-14 ***
## bdrms        15.19819     9.48352   1.603   0.113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 63.04 on 85 degrees of freedom
## Multiple R-squared:  0.6319, Adjusted R-squared:  0.6233
## F-statistic: 72.96 on 2 and 85 DF,  p-value: < 2.2e-16
```

```
coef_c2 <- coef(fit_c2)
se_c2 <- sqrt(diag(vcov(fit_c2)))
n_c2 <- nobs(fit_c2)
r2_c2 <- summary(fit_c2)$r.squared

list(coef = coef_c2, se = se_c2, n = n_c2, r2 = r2_c2)
```

```
## $coef
## (Intercept)      sqrft      bdrms
## -19.3149958    0.1284362  15.1981910
##
## $se
## (Intercept)      sqrft      bdrms
## 31.04661924    0.01382446  9.48351702
##
## $n
```

```
## [1] 88
##
## $r2
## [1] 0.6319184
```

```
cat(sprintf(
  "Estimated equation (SE in parentheses):\nprice = %.3f (%.3f) + %.5f (%.5f)*sqrft + %.3f (%.3f)*bdrms\n",
  coef_c2[1], se_c2[1],
  coef_c2[2], se_c2[2],
  coef_c2[3], se_c2[3],
  n_c2, r2_c2
))
```

Estimated equation (SE in parentheses):

price = -19.315 (31.047) + 0.12844 (0.01382)*sqrft* + 15.198 (9.484)*bdrms*

n = 88, R² = 0.6319

(ii) One more bedroom (holding *sqrft* constant)

```
delta_one_bed_thousands <- coef_c2["bdrms"]
delta_one_bed_dollars <- 1000 * delta_one_bed_thousands
delta_one_bed_dollars
```

```
##      bdrms
## 15198.19
```

(iii) One more bedroom + 140 square feet

```
delta_thousands <- 140 * coef_c2["sqrft"] + coef_c2["bdrms"]
delta_dollars <- 1000 * delta_thousands
delta_dollars
```

```
##      sqrft
## 33179.26
```

(iv) Percent variation explained (R-squared)

```
r2_c2
```

```
## [1] 0.6319184
```

(v) Predicted selling price for the first house

```
first_house <- hprice1[1, ]
pred_first_thousands <- predict(fit_c2, newdata = first_house)
pred_first_dollars <- 1000 * pred_first_thousands
pred_first_dollars
```

```
##      1
## 354605.2
```

(vi) Residual for the first house; underpaid or overpaid?

```
actual_first_thousands <- first_house$price
resid_first_thousands <- actual_first_thousands - pred_first_thousands
resid_first_dollars <- 1000 * resid_first_thousands

c(resid_thousands = resid_first_thousands, resid_dollars = resid_first_dollars)

## resid_thousands.1  resid_dollars.1
##           -54.60525           -54605.24898
```

Interpretation: - If residual < 0: actual price < predicted price → buyer paid less than predicted (underpaid relative to observables). - If residual > 0: buyer paid more than predicted (overpaid relative to observables).

3. Chapter 4 — Computer Exercise C1 (VOTE1)

3.1 What the problem is asking (plain English)

We estimate:

$$voteA = \beta_0 + \beta_1 \log(expendA) + \beta_2 \log(expendB) + \beta_3 prtystrA + u.$$

Tasks: 1. Interpret β_1 . 2. State the null hypothesis that a 1% increase in A's spending is offset by a 1% increase in B's spending. 3. Estimate the model and discuss whether A and B spending affect the outcome. 4. Estimate a model that directly gives the t statistic for testing the null in (2) (two-sided).

3.2 Step-by-step plan (how to solve it)

1) Fit the model with OLS: `lm(voteA ~ log(expendA) + log(expendB) + prtystrA)`.

2) Interpret log-level coefficients:

- A 1% increase in spending changes $\log(\text{spending})$ by about 0.01.
- So $\beta_1/100$ is approximately the percentage-point change in `voteA` from a 1% increase in `expendA`.

3) Offset null:

- “Offset” means the two 1% changes cancel:

$$\beta_1 + \beta_2 = 0.$$

4) Test properly:

- Use `linearHypothesis()` to test $\beta_1 + \beta_2 = 0$, or
- Reparameterize so one coefficient equals $\beta_1 + \beta_2$ and read its t statistic.

3.3 Common pitfalls

- **Log interpretation mistakes:** coefficients are tied to percentage changes, not dollar changes.
- **Testing incorrectly:** you need the standard error of $\hat{\beta}_1 + \hat{\beta}_2$.
- **Wrong-sided test:** part (iv) is two-sided.

3.4 Solution in R

```
data("vote1")
vote1 <- as_tibble(vote1)

vote1 %>% select(voteA, expendA, expendB, prtystA) %>% summary()
```

```
##      voteA      expendA      expendB      prtystA
## Min.   :16.0   Min.    : 0.302   Min.    : 0.93   Min.    :22.00
## 1st Qu.:36.0   1st Qu.: 81.634   1st Qu.: 60.05  1st Qu.:44.00
## Median :50.0   Median : 242.782   Median : 221.53  Median :50.00
## Mean   :50.5   Mean    : 310.611   Mean    : 305.09  Mean    :49.76
## 3rd Qu.:65.0   3rd Qu.: 457.410   3rd Qu.: 450.72  3rd Qu.:56.00
## Max.   :84.0   Max.    :1470.674   Max.    :1548.19  Max.    :71.00
```

```
fit_vote <- lm(voteA ~ log(expendA) + log(expendB) + prtystA, data = vote1)
summary(fit_vote)
```

```
##
## Call:
## lm(formula = voteA ~ log(expendA) + log(expendB) + prtystA,
##     data = vote1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.3968  -5.4174  -0.8679   4.9551  26.0660
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  45.07893    3.92631   11.48  <2e-16 ***
## log(expendA)   6.08332    0.38215   15.92  <2e-16 ***
## log(expendB)  -6.61542    0.37882  -17.46  <2e-16 ***
## prtystA        0.15196    0.06202    2.45  0.0153 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.712 on 169 degrees of freedom
## Multiple R-squared:  0.7926, Adjusted R-squared:  0.7889
## F-statistic: 215.2 on 3 and 169 DF, p-value: < 2.2e-16
```

(ii) State and test the offset null: $\beta_1 + \beta_2 = 0$

```
linearHypothesis(fit_vote, "log(expendA) + log(expendB) = 0")
```

```
## Linear hypothesis test
##
## Hypothesis:
## log(expendA) + log(expendB) = 0
##
## Model 1: restricted model
## Model 2: voteA ~ log(expendA) + log(expendB) + prtystA
##
##      Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1         170 10111
## 2         169 10052   1    59.261 0.9963 0.3196
```

(iv) Reparameterize to get the t statistic directly

Define: - $z_1 = \log(\text{expendA}) - \log(\text{expendB})$ - $z_2 = \log(\text{expendB})$

Then:

$$\beta_1 \log(\text{expendA}) + \beta_2 \log(\text{expendB}) = \beta_1 z_1 + (\beta_1 + \beta_2) z_2.$$

So the coefficient on z_2 equals $\beta_1 + \beta_2$.

```
fit_vote_reparam <- lm(
  voteA ~ I(log(expendA) - log(expendB)) + log(expendB) + prtysrA,
  data = vote1
)
```

```
summary(fit_vote_reparam)
```

```
##
## Call:
## lm(formula = voteA ~ I(log(expendA) - log(expendB)) + log(expendB) +
##     prtysrA, data = vote1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.3968  -5.4174  -0.8679   4.9551  26.0660
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    45.07893     3.92631  11.481  <2e-16 ***
## I(log(expendA) - log(expendB))  6.08332     0.38215  15.919  <2e-16 ***
## log(expendB)   -0.53210     0.53309  -0.998   0.3196
## prtysrA         0.15196     0.06202   2.450   0.0153 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.712 on 169 degrees of freedom
## Multiple R-squared:  0.7926, Adjusted R-squared:  0.7889
## F-statistic: 215.2 on 3 and 169 DF,  p-value: < 2.2e-16
```

Wrap-up

- **Ch.3 C2:** multiple regression, units, prediction, residual interpretation.
- **Ch.4 C1:** testing linear restrictions correctly (offset hypothesis).
- **Tools:** VIF (diagnostic), `linearHypothesis()` (clean restriction tests).